Computer Networks

Interview Preparation Guide

Top 20 Most Asked Questions & Answers
Intermediate to Advanced Level | With Easy Explanations

Level	Count	Topics Covered
Intermediate	12	OSI Model, TCP/UDP, 3-Way Handshake, IP Addressing, DNS, HTTP/HTTPS, Switch/Router/Hub, Subnetting, Firewall, Client-Server
Advanced	8	TCP Congestion Control, ARP, NAT, Load Balancing, VPN, Routing Protocols, SSL/TLS Handshake, Proxy Servers

Structured for quick revision before technical interviews

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Q1 **OSI Model** Intermediate

What is the OSI Model? Name its 7 layers.

ANSWER

The OSI (Open Systems Interconnection) Model is a conceptual 7-layer framework that standardizes how data travels from one device to another over a network. Each layer has a specific job and communicates only with the layers directly above and below it.

The 7 layers (top to bottom) are: 7. Application, 6. Presentation, 5. Session, 4. Transport, 3. Network, 2. Data Link, 1. Physical. A common mnemonic is 'All People Seem To Need Data Processing'.

EASY EXPLANATION: Think of sending a birthday gift through a courier company: Application = deciding what gift to buy, Presentation = wrapping it nicely, Session = keeping the order tracking active, Transport = choosing express or standard delivery, Network = picking the best route, Data Link = the local delivery van, Physical = the actual road the van drives on.

Q2 **TCP vs UDP** Intermediate

What is the difference between TCP and UDP?

ANSWER

TCP (Transmission Control Protocol) is connection-oriented — it establishes a connection first (via a 3-way handshake), guarantees reliable, ordered delivery of data, and has error checking with retransmission of lost packets. It's slower due to this overhead. Used for web browsing (HTTP), email, file transfer.

UDP (User Datagram Protocol) is connectionless — it sends data (datagrams) without establishing a connection first, does not guarantee delivery, order, or error correction, but is much faster with lower overhead. Used for video streaming, online gaming, DNS, and VoIP where speed matters more than perfect accuracy.

EASY EXPLANATION: TCP is like sending a registered post letter — you get confirmation it was delivered, and if it's lost, it's resent. UDP is like shouting a message across a room — it's fast, but you're not 100% sure everyone heard every word, and you don't wait to check.

Q3 **3-Way Handshake** Intermediate

Explain the TCP 3-Way Handshake.

ANSWER

The 3-way handshake is how TCP establishes a reliable connection between a client and server before any real data is sent:

1. SYN - Client sends a SYN (synchronize) packet to the server, requesting a connection and proposing an initial sequence number.

2. SYN-ACK - Server responds with a SYN-ACK packet, acknowledging the client's request and sending its own sequence number.

3. ACK - Client sends back an ACK (acknowledgment) packet, confirming receipt. The connection is now established and data transfer can begin.

EASY EXPLANATION: It's like a phone call: You dial (SYN), the other person picks up and says 'Hello, can you hear me?' (SYN-ACK), and you reply 'Yes, I can hear you' (ACK). Only after this exchange does the real conversation (data) start.

Q4 **IP Addressing** Intermediate

What is an IP Address, and what is the difference between IPv4 and IPv6?

ANSWER

An IP address is a unique numerical label assigned to every device on a network, used to identify and locate it so data can be routed correctly.

IPv4 uses a 32-bit address (e.g., 192.168.1.1), giving about 4.3 billion unique addresses — which have essentially run out given the number of connected devices today.

IPv6 uses a 128-bit address (e.g., 2001:0db8::1), providing a vastly larger address space (essentially unlimited for practical purposes), along with built-in simplification for auto-configuration and security.

EASY EXPLANATION: An IP address is like a home address for your device on the internet — it's how data knows where to go. IPv4 is like an old numbering system for houses that's running out of numbers as new houses (devices) keep being built; IPv6 is the redesigned system with practically infinite house numbers available.

Q5 **DNS** Intermediate

What is DNS and how does DNS Resolution work?

ANSWER

DNS (Domain Name System) is often called the 'phonebook of the internet' — it translates human-readable domain names (like google.com) into machine-readable IP addresses (like 142.250.190.14).

Resolution steps: your browser first checks its local cache; if not found, it asks a Recursive Resolver (usually from your ISP), which queries a Root DNS Server, then a TLD (Top Level Domain) Server (for .com, .org, etc.), and finally the Authoritative Name Server, which returns the actual IP address.

EASY EXPLANATION: You don't remember your friend's phone number by heart — you look up their name in your contacts app, which returns the number to dial. DNS does exactly that for websites: you type a name, and DNS looks up the corresponding number (IP address) to actually connect to.

Q6 **HTTP vs HTTPS** Intermediate

What is the difference between HTTP and HTTPS?

ANSWER

HTTP (HyperText Transfer Protocol) is the standard protocol for transferring web data between a client (browser) and a server, but the data is sent in plain text — anyone intercepting it can read the content.

HTTPS (HTTP Secure) adds a layer of encryption using SSL/TLS on top of HTTP. It encrypts data in transit, verifies the server's identity through certificates, and ensures data integrity — preventing eavesdropping and tampering.

EASY EXPLANATION: HTTP is like sending a postcard — anyone who handles it along the way can read what's written on it. HTTPS is like sending the same message in a sealed, tamper-proof envelope that only the intended recipient can open and read.

Q7 **Switch vs Router vs Hub** Intermediate

What is the difference between a Hub, Switch, and Router?

ANSWER

A Hub is a basic Layer 1 (Physical) device that broadcasts incoming data to ALL connected devices, regardless of the intended recipient — inefficient and now largely obsolete.

A Switch is a Layer 2 (Data Link) device that uses MAC addresses to send data only to the specific device it's intended for within a local network (LAN), making it far more efficient than a hub.

A Router is a Layer 3 (Network) device that connects different networks together (e.g., your home network to the internet) and uses IP addresses to determine the best path to forward data between networks.

EASY EXPLANATION: A Hub is like shouting an announcement to an entire office so everyone hears it, even if it's meant for one person. A Switch is like handing a note directly to the one person it's addressed to within the office. A Router is like the mailroom that decides how to send a letter to a completely different building (network) altogether.

Q8 **Subnetting** Intermediate

What is Subnetting and why is it used?

ANSWER

Subnetting is the process of dividing a large network into smaller, more manageable sub-networks (subnets), done by borrowing bits from the host portion of an IP address to create a subnet mask.

Benefits: improves network performance by reducing broadcast traffic, enhances security by isolating segments, and makes more efficient use of the available IP address space by allocating only as many addresses as each subnet actually needs.

EASY EXPLANATION: Imagine a huge office building (one big network) with no separate floors — everyone hears every announcement and it's chaotic. Subnetting is like dividing that building into separate floors/departments (subnets), each with its own internal announcements, making things organized, secure, and efficient.

Q9 **Application Layer Protocols** Intermediate

Name common Application Layer protocols and their uses.

ANSWER

HTTP/HTTPS - used for transferring web pages between browsers and servers.

FTP (File Transfer Protocol) - used for transferring files between a client and server.

SMTP (Simple Mail Transfer Protocol) - used for sending emails.

POP3/IMAP - used for retrieving/managing emails from a mail server.

DNS - used for resolving domain names to IP addresses.

DHCP (Dynamic Host Configuration Protocol) - automatically assigns IP addresses to devices on a network.

EASY EXPLANATION: These are like different specialized departments in a company, each handling one specific job: one handles web pages, one handles files, one handles mail delivery, one handles mail pickup, one handles 'phonebook' lookups, and one hands out ID badges (IP addresses) to new employees (devices) automatically.

Q10 **Firewall** Intermediate

What is a Firewall and how does it work?

ANSWER

A firewall is a network security device/software that monitors and controls incoming and outgoing network traffic based on a defined set of security rules, acting as a barrier between a trusted internal network and untrusted external networks (like the internet).

It works by inspecting packets against rules (based on IP address, port number, protocol) and either allowing or blocking them. Types include packet-filtering firewalls, stateful inspection firewalls, and application-layer (proxy) firewalls.

EASY EXPLANATION: A firewall is like a security guard at a building entrance who checks everyone's ID against a list before letting them in or out — legitimate visitors (allowed traffic) pass through, while suspicious or unauthorized visitors (malicious traffic) are stopped at the door.

Q11 **Client-Server Model** Intermediate

What is the Client-Server Model?

ANSWER

The client-server model is a network architecture where tasks are divided between service providers (servers) and service requesters (clients). The client sends a request over the network, and the server processes it and sends back a response.

Servers are typically powerful, always-on machines that manage resources (databases, files, web pages) and can serve multiple clients simultaneously, while clients are the end-user devices making requests.

EASY EXPLANATION: It's like a restaurant: you (the client) don't cook your own food — you place an order (request) with the waiter, the kitchen (server) prepares it, and the waiter brings back your food

(response). Many customers can order from the same kitchen at once.

Q12 **Congestion Control** Intermediate

What is Congestion Control in networks?

ANSWER

Congestion control refers to techniques used to prevent and manage network congestion — a situation where too much data is being sent through a network, overwhelming routers/links, causing packet loss and delays.

TCP handles this using algorithms like Slow Start (gradually increasing the sending rate), Congestion Avoidance (linear increase after a threshold), and Fast Retransmit/Recovery (quickly reacting to detected packet loss by reducing the sending rate).

EASY EXPLANATION: Think of it like traffic entering a highway: if everyone floods in at once, there's a massive jam. Congestion control is like a metered on-ramp light that lets cars (data) in gradually, speeding up the flow only as long as the highway (network) can handle it smoothly.

Advanced Level Questions

Q13	TCP Congestion Control Algorithms	Advanced

Explain TCP's Slow Start and Congestion Avoidance phases in detail.

ANSWER

Slow Start: TCP begins with a small congestion window (cwnd), typically 1 MSS (Maximum Segment Size), and DOUBLES it every round-trip time (exponential growth) as long as acknowledgments are received successfully. This continues until cwnd reaches a threshold called ssthresh.

Congestion Avoidance: once cwnd crosses ssthresh, growth switches from exponential to linear (increasing by roughly 1 MSS per round-trip time) to probe for available bandwidth more cautiously and avoid triggering congestion.

If packet loss is detected, ssthresh is reduced (often halved) and cwnd resets — either to 1 (classic Slow Start restart, on timeout) or to the new ssthresh (Fast Recovery, on triple duplicate ACKs).

EASY EXPLANATION: It's like slowly test-driving how fast you can push a shopping cart through a crowded store: you start slow and speed up quickly at first (Slow Start). Once you sense it's getting risky, you speed up much more cautiously (Congestion Avoidance). If you bump into someone (packet loss), you immediately slow way down again and restart your cautious acceleration.

Q14	ARP	Advanced

What is ARP (Address Resolution Protocol) and how does it work?

ANSWER

ARP is a protocol used to map an IP address (Layer 3 / logical address) to a MAC address (Layer 2 / physical hardware address) within a local network, since devices actually communicate at the hardware level using MAC addresses.

Process: when a device wants to send data to an IP address on its local network but doesn't know the corresponding MAC address, it broadcasts an ARP Request ('Who has this IP address?') to all devices on the LAN. The device owning that IP replies with an ARP Reply containing its MAC address, which the sender then caches for future use (ARP cache/table).

EASY EXPLANATION: Imagine you know someone's apartment number (IP address) in a building but not which specific door/mailbox (MAC address) it is. You shout out in the hallway 'Whoever lives in Apartment 5, please tell me your exact door number!' and that person responds directly. Now you know exactly which door to knock on next time.

Q15	Network Address Translation (NAT)	Advanced

What is NAT and why is it needed?

ANSWER

NAT (Network Address Translation) is a technique that allows multiple devices on a private/local network (using private IP addresses) to share a single public IP address when communicating with the

internet. The router translates private IPs to the public IP (and vice versa for responses) and tracks each connection using port numbers.

It's needed primarily because of IPv4 address exhaustion — there aren't enough public IPv4 addresses for every single device in the world, so NAT lets an entire household or organization share just one public address. It also adds a layer of security by hiding internal private IP addresses from the outside internet.

EASY EXPLANATION: It's like an office building having one official street address (public IP) even though there are hundreds of employees (devices) inside with their own desk/cubicle numbers (private IPs). Mail (data) sent to the building gets routed internally by the receptionist (router/NAT) to the correct desk based on tracked records, without outsiders needing to know each individual desk number.

Q16 **Load Balancing** Advanced

What is Load Balancing, and what are common algorithms used?

ANSWER

Load balancing is the process of distributing incoming network traffic or requests across multiple servers so that no single server becomes overwhelmed, improving availability, reliability, and response time.

Common algorithms: Round Robin (requests distributed sequentially across servers in rotation), Least Connections (send to the server currently handling the fewest active connections), IP Hash (route based on a hash of the client's IP, ensuring the same client consistently reaches the same server), and Weighted Round Robin (servers with higher capacity get proportionally more requests).

EASY EXPLANATION: It's like a bank with multiple teller counters and a manager directing the queue. Instead of everyone rushing to one teller (server) while others sit idle, the manager (load balancer) sends each new customer to whichever counter is free or least busy, so everyone gets served faster overall.

Q17 **VPN** Advanced

What is a VPN and how does it ensure secure communication?

ANSWER

A VPN (Virtual Private Network) creates a secure, encrypted 'tunnel' between a user's device and a remote server over the public internet, making it appear as though the device is directly connected to a private network — even though the underlying data actually travels over shared public infrastructure.

Security is achieved through tunneling protocols (like OpenVPN, WireGuard, or IPsec) that encrypt data before it leaves the device and decrypt it only at the VPN server, along with authentication to verify both ends of the connection, protecting data from eavesdropping and masking the user's real IP address.

EASY EXPLANATION: It's like traveling through a private, opaque tunnel underneath a busy public highway, instead of driving in the open where everyone can see your car and where you're going. Even though you're using the same general route (the internet), no one on the outside can see what you're doing inside that tunnel.

Q18	Distance Vector vs Link State Routing	Advanced

What is the difference between Distance Vector and Link State Routing protocols?

ANSWER

Distance Vector routing (e.g., RIP) works by each router sharing its ENTIRE routing table (destination + distance/hop count) periodically with its directly connected neighbors only. Each router builds its routing table based only on what neighbors tell it — it doesn't know the full network topology. This is simple but converges slowly and can face issues like routing loops (mitigated by techniques like split horizon).

Link State routing (e.g., OSPF) works by each router gathering information about its directly connected links and FLOODING this information to ALL routers in the network (not just neighbors). Every router then independently builds a complete map of the network topology and calculates the shortest path using Dijkstra's algorithm. This converges faster and scales better but requires more processing and memory.

EASY EXPLANATION: Distance Vector is like asking your immediate neighbors 'how far is the nearest hospital from your house?' and trusting their answer without seeing the full city map yourself. Link State is like every person getting a complete map of the entire city and calculating the best route to the hospital themselves — more accurate and faster to adapt to road closures, but needs more information upfront.

Q19	SSL/TLS Handshake	Advanced

Explain how the SSL/TLS Handshake establishes a secure connection.

ANSWER

The TLS handshake happens after the TCP 3-way handshake and establishes an encrypted session between client and server:

1. Client Hello - client sends supported TLS versions, cipher suites, and a random number.

2. Server Hello - server picks a cipher suite, sends its digital certificate (containing its public key), and its own random number.

3. Verification & Key Exchange - the client verifies the server's certificate against trusted Certificate Authorities, then both sides use asymmetric encryption (public/private key pairs) to securely agree on a shared 'session key'.

4. Secure Communication - once both sides have the same session key, they switch to fast symmetric encryption using that key for the rest of the actual data transfer.

EASY EXPLANATION: It's like two people who've never met agreeing on a secret code before whispering (client/server hello and cipher agreement), one showing an ID card to prove who they are (certificate verification), then briefly using a complex secret handshake to agree on a much simpler shared password (key exchange) — after which they just use that simple shared password (symmetric session key) for fast, secure conversation from then on.

Q20	Proxy Server	Advanced

What is a Proxy Server, and what is the difference between Forward and Reverse Proxy?

ANSWER

A proxy server acts as an intermediary between a client and the destination server, forwarding requests and responses on behalf of one side, often adding caching, filtering, anonymity, or load distribution.

A Forward Proxy sits in front of CLIENTS — it forwards client requests OUT to the internet, often used to hide client identity, enforce content filtering/access policies, or cache frequently requested content (common in corporate networks or to bypass geo-restrictions).

A Reverse Proxy sits in front of SERVERS — it receives incoming requests from the internet and forwards them to the appropriate backend server, commonly used for load balancing, SSL termination, caching, and hiding the internal server architecture from the public internet.

EASY EXPLANATION: A Forward Proxy is like a personal assistant who makes calls to outsiders on YOUR behalf, so the outsiders only ever see the assistant's number, not yours. A Reverse Proxy is like a company receptionist who takes all incoming calls FOR the company and routes each one to the correct internal department, so outside callers never directly reach or even know about the individual employees (backend servers).